

$i^{th}$  MLP Layer ( $\forall i \in \{1, 2, \dots, L\}$ )

$\mathbf{h}^i \in \mathbb{R}^{T \times d}$   
 $h_j^i$ :  $j^{th}$  Neuron of  $i^{th}$  Layer  
 $(\forall j \in \{1, 2, \dots, 4d\})$

Non-Linear Activation  
 $SiLU(\cdot)$

Gate Projection  
 $\mathbf{W}_1^i \in \mathbb{R}^{d \times 4d}$

Self-Attention  
 $\mathbf{W}_q^i, \mathbf{W}_k^i, \mathbf{W}_v^i, \mathbf{W}_o^i$

Input to  $i^{th}$  Layer  
 $\mathbf{H}^{i-1} \in \mathbb{R}^{T \times d}$

Elementwise Mult  
 $\otimes$

Down Projection  
 $\mathbf{W}_2^i \in \mathbb{R}^{4d \times d}$

Output to  $i^{th}$  Layer  
 $\mathbf{H}^i \in \mathbb{R}^{T \times d}$

Up Projection  
 $\mathbf{W}_3^i \in \mathbb{R}^{d \times 4d}$

